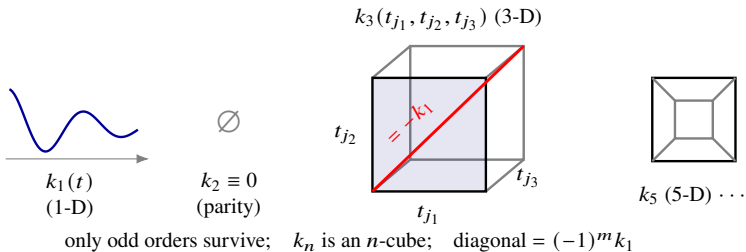


A. Kernel tensor — only odd orders survive

each tensor entry from n signed virtual-Z kicks $Z_{s_a}(t_{j_a})$ at the scanned times:

$$\widehat{k}_n(t_{j_1}, \dots, t_{j_n}) = \frac{1}{(2\phi_z)^n} \sum_{s \in \{\pm 1\}^n} \left(\prod_a s_a \right) q_s, \quad q_s = \frac{p_x - p_{-x}}{2}$$



B. Transient-field calibration

$$G_\alpha^{(n)} = \underbrace{\int \dots \int}_n k_n(t_1, \dots, t_n) d^n t$$

integrate →

$$p_{\text{diff}}(\Delta) = G_1 \Delta + \frac{1}{6} G_3 \Delta^3 + \dots$$

(only odd powers)

→ invert transient detuning $\delta\omega(t)$

Two routes to the tensor entries. *Experiment:* alternating-sign scan, cost $2^n M^n N_{\text{shot}}$ shots. *Simulation:* one evolution $U(t)$, then $k_n = i^n \langle 0 | [W(t_n), [\dots, [W(t_1), Q]]] | 0 \rangle$ with $W = U^\dagger \sigma_z U / 2$ — the whole tensor from a single run.